Assignment-6

1. Display EMP_no, E_name from EMP using cursor?

Ans:-

<u>Code:-</u>

```
SET SERVEROUTPUT ON;

DECLARE

   CURSOR c_emp IS

      SELECT empno, ename FROM emp;



   v_empno emp.empno%TYPE;

   v_ename emp.ename%TYPE;

BEGIN

   OPEN c_emp;

   LOOP

      FETCH c_emp INTO v_empno, v_ename;

      EXIT WHEN c_emp%NOTFOUND;



      DBMS_OUTPUT.PUT_LINE('EMP NO: ' || v_empno ||

                        '  NAME: ' || v_ename);

   END LOOP;

   CLOSE c_emp;

END;

/
```

Output:-

```
SQL Plus

SQL> SET SERVEROUTPUT ON;
SQL> DECLARE
  2      CURSOR c_emp IS
  3          SELECT empno, ename FROM emp;
  4
  5      v_empno emp.empno%TYPE;
  6      v_ename emp.ename%TYPE;
  7  BEGIN
  8      OPEN c_emp;
  9      LOOP
 10          FETCH c_emp INTO v_empno, v_ename;
 11          EXIT WHEN c_emp%NOTFOUND;
 12
 13          DBMS_OUTPUT.PUT_LINE('EMP NO: ' || v_empno ||
 14                               '  NAME: ' || v_ename);
 15      END LOOP;
 16      CLOSE c_emp;
 17  END;
 18  /
EMP NO: 7369  NAME: SMITH
EMP NO: 7499  NAME: ALLEN
EMP NO: 7521  NAME: WARD
EMP NO: 7566  NAME: JONES
EMP NO: 7654  NAME: MARTIN
EMP NO: 7698  NAME: BLAKE
EMP NO: 7782  NAME: CLERK
EMP NO: 7788  NAME: SCOTT
EMP NO: 7839  NAME: KING
EMP NO: 7844  NAME: TURNER
EMP NO: 7876  NAME: ADAMS
EMP NO: 7900  NAME: JAMES
EMP NO: 7902  NAME: FORD
EMP NO: 7934  NAME: MILLER

PL/SQL procedure successfully completed.
```

2. Write a cursor to find the first five employees getting the highest salary?

Ans:-

Code:-

```
SET SERVEROUTPUT ON;

DECLARE

   CURSOR c_sal IS

      SELECT empno, ename, sal

      FROM emp
```

```
        ORDER BY sal DESC;


    v_empno emp.empno%TYPE;
    v_ename emp.ename%TYPE;
    v_sal   emp.sal%TYPE;
    cnt NUMBER := 0;
BEGIN
    OPEN c_sal;
    LOOP
        FETCH c_sal INTO v_empno, v_ename, v_sal;
        EXIT WHEN c_sal%NOTFOUND OR cnt = 5;


        cnt := cnt + 1;
        DBMS_OUTPUT.PUT_LINE(v_empno || '  ' ||
                        v_ename || '  ' || v_sal);
    END LOOP;
    CLOSE c_sal;
END;
/
```

Output:-

```
SQL Plus

SQL> SET SERVEROUTPUT ON;
SQL> DECLARE
  2      CURSOR c_sal IS
  3          SELECT empno, ename, sal
  4          FROM emp
  5          ORDER BY sal DESC;
  6
  7      v_empno emp.empno%TYPE;
  8      v_ename emp.ename%TYPE;
  9      v_sal   emp.sal%TYPE;
 10      cnt NUMBER := 0;
 11  BEGIN
 12      OPEN c_sal;
 13      LOOP
 14          FETCH c_sal INTO v_empno, v_ename, v_sal;
 15          EXIT WHEN c_sal%NOTFOUND OR cnt = 5;
 16
 17          cnt := cnt + 1;
 18          DBMS_OUTPUT.PUT_LINE(v_empno || '  ' ||
 19                               v_ename || '  ' || v_sal);
 20      END LOOP;
 21      CLOSE c_sal;
 22  END;
 23  /
7839  KING  5000
7902  FORD  3000
7788  SCOTT  3000
7566  JONES  2975
7698  BLAKE  2850

PL/SQL procedure successfully completed.
```

3. Write a cursor to calculate DA and store in the DAP?

Ans:-

Code:-

First create a table named as DAP using the following code:-

```
CREATE TABLE dap (
   emp_no NUMBER,
   da     NUMBER
);
```

After creating the table then applies this code:-

```
SET SERVEROUTPUT ON;

DECLARE

   CURSOR c_da IS

      SELECT empno, sal FROM emp;


   v_empno emp.empno%TYPE;

   v_sal   emp.sal%TYPE;

   v_da    NUMBER;

BEGIN

   OPEN c_da;

   LOOP

      FETCH c_da INTO v_empno, v_sal;

      EXIT WHEN c_da%NOTFOUND;


      v_da := v_sal * 0.30;


      INSERT INTO dap(emp_no, da)

      VALUES (v_empno, v_da);

   END LOOP;


   CLOSE c_da;

   COMMIT;
```

END;

/

Output:-

```
SQL Plus

SQL> CREATE TABLE dap (
  2      emp_no NUMBER,
  3      da     NUMBER
  4  );

Table created.

SQL> SET SERVEROUTPUT ON;
SQL> DECLARE
  2      CURSOR c_da IS
  3          SELECT empno, sal FROM emp;
  4
  5      v_empno emp.empno%TYPE;
  6      v_sal   emp.sal%TYPE;
  7      v_da    NUMBER;
  8  BEGIN
  9      OPEN c_da;
 10      LOOP
 11          FETCH c_da INTO v_empno, v_sal;
 12          EXIT WHEN c_da%NOTFOUND;
 13
 14          v_da := v_sal * 0.30;
 15
 16          INSERT INTO dap(emp_no, da)
 17          VALUES (v_empno, v_da);
 18      END LOOP;
 19
 20      CLOSE c_da;
 21      COMMIT;
 22  END;
 23  /

PL/SQL procedure successfully completed.
```

```
SQL> select * from dap;

    EMP_NO         DA
---------- ----------
      7369        240
      7499        480
      7521        375
      7566      892.5
      7654        375
      7698        855
      7782        735
      7788        900
      7839       1500
      7844        450
      7876        330

    EMP_NO         DA
---------- ----------
      7900        285
      7902        900
      7934        390

14 rows selected.
```

4. Write a program to display all classes from the EMP table using cursor?

Ans:-

<u>Code:-</u>

```
SET SERVEROUTPUT ON;

DECLARE

   CURSOR c_dept IS

      SELECT DISTINCT deptno FROM emp;



   v_dept emp.deptno%TYPE;
```

```
BEGIN
   OPEN c_dept;
   LOOP
      FETCH c_dept INTO v_dept;
      EXIT WHEN c_dept%NOTFOUND;


      DBMS_OUTPUT.PUT_LINE('DEPT NO: ' || v_dept);
   END LOOP;
   CLOSE c_dept;
END;
/
```

Output:-

```
SQL Plus

SQL> SET SERVEROUTPUT ON;
SQL> DECLARE
  2      CURSOR c_dept IS
  3          SELECT DISTINCT deptno FROM emp;
  4
  5      v_dept emp.deptno%TYPE;
  6  BEGIN
  7      OPEN c_dept;
  8      LOOP
  9          FETCH c_dept INTO v_dept;
 10          EXIT WHEN c_dept%NOTFOUND;
 11
 12          DBMS_OUTPUT.PUT_LINE('DEPT NO: ' || v_dept);
 13      END LOOP;
 14      CLOSE c_dept;
 15  END;
 16  /
DEPT NO: 20
DEPT NO: 30
DEPT NO: 10

PL/SQL procedure successfully completed.
```